Constructive and Destructive Training Regimes: Gradient Diagnostics for Neural Network Stability

Thiago Mattar, Cristiano L. Castro, Frederico Coelho, Yan G. Ferreira, Antonio P. Braga

***Abstract*—Destructive interactions between successive gradients can accompany sustained training progress, yet strong gradient amplification can turn this behavior into sharp loss excursions. We develop a diagnostic framework that distinguishes constructive and destructive regimes through adjacent-gradient cosine and interprets them jointly with gradient magnitude and cumulative parameter displacement. An exact SGD displacement decomposition and local loss expansions connect these measurements to temporal interference, directional curvature and sampling noise. The diagnostics reuse training gradients with linear-time vector operations. Controlled CIFAR-10 and WikiText-2 experiments characterize productive destructive regimes under full-batch GD and their response to mini-batch sampling. Under Adam, raw-gradient dynamics remain informative along adaptive trajectories and reveal early signs of developing loss spikes. Across 129 recovered spikes larger than 10%, gradient norms increase in every event, and 117 of 127 defined pre-rise cosines identify a destructive regime. Joint measures of pre-rise alignment and rise norm amplification explain 38–94% of the within-run variation in log relative peak loss. These results connect inexpensive online measurements to the anatomy of local loss instability and motivate diagnostic-guided interventions.**

***Index Terms*—Neural Network Optimization, Training Stability, Gradient Alignment, Stochastic Gradient Descent, Adam**

I. Introduction

Neural networks often reduce training loss while successive gradients point in nearly opposite directions. This behavior becomes especially visible at large learning rates, where rapid overall progress can coexist with oscillation and sharp loss spikes [1]–[3]. For practitioners, the important question is when such oscillation remains productive and when it develops into a damaging excursion. Answering it requires diagnostics that expose the changing training dynamics early enough to inform a response.

We approach this question through *constructive and destructive regimes*. The cosine similarity ρ_t between adjacent raw gradients identifies the regime: positive values indicate directional reinforcement, and negative values indicate directional opposition. The gradient norm supplies the complementary information about the strength of that interaction. Together, they distinguish persistent reversal during descent from reversal accompanied by rapid gradient amplification. This distinction is central to the paper: a destructive regime can sustain loss reduction, while sufficiently strong amplification can turn its local oscillations into loss-increasing steps.

Classical stability analysis relates step size to the Hessian spectrum, and stochastic extensions incorporate variation induced by sampling [4], [5]. The Edge-of-Stability literature establishes that deep networks can continue to improve while operating near or beyond the local quadratic stability boundary [1], [6], [7]. Gradient-product diagnostics and hypergradient methods already exploit successive directions to detect optimization phases or adapt the learning rate [8]–[10]. Our contribution is to connect these temporal interactions to a common account of regime, gradient amplification, net parameter motion and loss-spike anatomy, using measurements available along the training trajectory.

The formulation starts from an exact decomposition of SGD displacement into individual step magnitudes and interactions across all temporal lags. Adjacent-gradient cosine describes the current regime; normalized displacement measures how much traveled distance survives as net motion. A local loss expansion then shows why destructive steps can either decrease or increase loss, depending on gradient-norm growth. The projected-change diagnostic $\eta\widehat{\kappa}_t$ summarizes this interaction and connects it to directional curvature. These quantities require linear-time vector operations on gradients already computed by training, making them suitable for online monitoring.

We first characterize the regimes in controlled full-batch GD and mini-batch SGD experiments on CIFAR-10 and WikiText-2. We then extend the analysis to Adam, whose adaptive directions transform the raw gradient before each update [11]. Raw-gradient diagnostics still expose the loss landscape encountered along that adaptive path: the largest spikes show diagnostic changes before the loss maximum, and the Transformer example reveals a warning before its pronounced rise. Across 129 recovered spikes larger than 10%, gradient norms grow in every event and 117 of the 127 defined pre-rise cosines identify a destructive regime. Combining that initial regime with subsequent norm amplification accounts for a substantial fraction of the variation in spike magnitude.

The paper makes three contributions:

- A unified diagnostic formulation linking constructive and destructive regimes, cumulative parameter displacement and local loss changes, with explicit curvature and sampling-noise terms.

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- A controlled empirical characterization showing how destructive regimes sustain descent, reduce displacement retention and change under mini-batch sampling.
- An extension to Adam that identifies early signs and recurring anatomy of loss spikes through raw-gradient alignment and amplification, supported by individual events, amplitude distributions and quantitative associations.

Section II positions the diagnostics within stability and optimization research. Section III develops the formulation, and Section IV characterizes the regimes under GD and SGD. Section V examines their role in Adam loss spikes and motivates diagnostic-guided interventions. Section VI concludes.

II. Background and Related Diagnostics

Training dynamics reflect both the geometry encountered by an update and its interaction with preceding updates. Curvature-based analyses explain local amplification; stochastic analyses add sampling variation; temporal diagnostics describe how successive directions reinforce or oppose one another. We use these perspectives to develop a practical account of constructive and destructive regimes.

A. Curvature, Oscillation and Stability

Let $\theta_t \in \mathbb{R}^p$ denote the parameters after update t and g_t the gradient evaluated at θ_{t-1}. For a constant learning rate $\eta > 0$, plain SGD follows

$$\theta_t = \theta_{t-1} - \eta g_t. \tag{1}$$

In full-batch GD, $g_t = \nabla\mathcal{L}(\theta_{t-1})$ for the training objective $\mathcal{L}$. Writing $H_t = \nabla^2\mathcal{L}(\theta_{t-1})$, the deterministic update has Jacobian

$$J_t = I - \eta H_t. \tag{2}$$

This matrix describes the response to a small parameter perturbation. For a fixed positive-definite quadratic Hessian, each eigencomponent is multiplied by $1 - \eta\lambda_i$, and linear stability requires

$$\max_i |1 - \eta\lambda_i| < 1 \quad \Longleftrightarrow \quad \eta\lambda_{\max}(H) < 2. \tag{3}$$

The component contracts without changing sign when $0 < \eta\lambda_i < 1$, alternates while contracting when $1 < \eta\lambda_i < 2$, and alternates with increasing magnitude when $\eta\lambda_i > 2$ [4], [12]. Thus oscillation and amplification are distinct features even in the simplest local model.

Deep-network trajectories repeatedly enter oscillatory regions. Cohen et al. [1] show that full-batch GD often approaches $\lambda_{\max} \simeq 2/\eta$ while reducing loss over longer timescales. Analyses of this Edge-of-Stability behavior explain how nonlinear dynamics sustain progress near the quadratic boundary [6], [7]. The trajectory measurements of Xing et al. [13] likewise describe steps crossing between sides of a valley. These results motivate examining the gradients produced by successive crossings: their alignment reveals reversal, and their magnitudes reveal whether that reversal is shrinking or growing.

B. Sampling Noise and Local Geometry

Mini-batch sampling changes both the estimated direction and the trajectory followed through the objective [14]–[16]. Under fresh, conditionally unbiased sampling, write

$$\begin{aligned} g_t &= \nabla\mathcal{L}(\theta_{t-1}) + \xi_t, \\ \mathbb{E}[\xi_t \mid \theta_{t-1}] &= 0, \\ \mathrm{Cov}(\xi_t \mid \theta_{t-1}) &= \frac{N-b}{b(N-1)}\Sigma(\theta_{t-1}), \end{aligned} \tag{4}$$

where the last equality assumes a uniformly sampled size-b subset of $N > 1$ examples without replacement. At a fixed state, let $h_i = \nabla\ell_i(\theta)$ and $\mu = N^{-1}\sum_i h_i = \nabla\mathcal{L}(\theta)$. The per-example covariance is

$$\Sigma(\theta) = \frac{1}{N}\sum_{i=1}^{N}(h_i - \mu)(h_i - \mu)^\top. \tag{5}$$

Its trace measures the total squared dispersion around the full gradient. The finite-population factor in Eq. (4) shows how larger batches reduce this dispersion.

Curvature and stochasticity also interact. The break-even analysis of Jastrzebski et al. [5] links the early learning rate and batch size to the curvature and gradient variability encountered later. In its local one-dimensional quadratic model, variation in sampled curvature adds to the mean-square amplification produced by the mean curvature. This gives a reason to examine the regime along the evolving path as well as the sampling variance at a fixed point. Section III-B makes that distinction explicit for adjacent-gradient alignment.

C. Temporal Gradient Diagnostics

Successive gradient products have a substantial history in optimization diagnostics. Chee and Toulis [8] use them to detect the transition toward a stationary phase of constant-step SGD. Pesme et al. [9] analyze the classical gradient-product test and show how its small signal relative to variation can impair phase detection. Baydin et al. [10] derive online learning-rate adaptation from hypergradients; for ordinary gradient descent, the resulting update uses the inner product of successive gradients.

Our focus is the anatomy of the regime that these temporal interactions reveal. We combine adjacent alignment with gradient amplification and cumulative displacement, then connect those measurements to loss behavior. This joint view explains how destructive regimes support sustained descent and identifies the additional amplification associated with large loss excursions. Extending the same measurements to raw gradients under Adam provides a common interpretation across plain and adaptive updates.

D. Loss Spikes in Practical Training

Large-model training reports show why early diagnostics matter. PaLM describes approximately 20 loss spikes during 540B-model training and recovers by restarting from an earlier checkpoint and skipping batches [2]. OPT reports growing activation norms and loss divergence, with recovery through checkpoint restarts and learning-rate reduction; tighter gradient

clipping also improves early stability [17]. These interventions incur training cost and make the timing of a warning practically consequential.

Modern recipes therefore monitor gradient norms alongside loss. OLMo 2 uses rolling loss and gradient-norm spike scores in its stability analysis [3], while Llama 3 emphasizes numerical precision, warmup and batch-size scheduling in a largely stable pretraining recipe [18]. Adjacent-gradient cosine adds a directional interpretation to such magnitude measurements: it identifies whether a growing gradient reinforces or opposes the preceding one. The following section develops this combination into diagnostics of constructive and destructive training.

III. Characterizing Constructive and Destructive Regimes

Our diagnostics answer three connected questions: which directional regime is the optimizer traversing, how strongly are gradients changing, and how much of the resulting motion accumulates? Adjacent-gradient cosine ρ_t, the projected change $\eta\widehat{\kappa}_t$, and normalized displacement R_T supply these local and cumulative views. We first derive them for plain SGD, then explain how destructive regimes can sustain descent or develop loss-increasing steps.

Consider the plain SGD update in Eq. (1), with constant learning rate $\eta > 0$. As before, $\theta_t \in \mathbb{R}^p$ denotes the parameters after update t, and g_t is the mean mini-batch gradient evaluated at θ_{t-1}. Summing the updates gives the displacement from initialization,

$$\Delta_T := \theta_T - \theta_0 = -\eta \sum_{t=1}^{T} g_t. \tag{6}$$

The displacement is the net parameter change produced by T updates. Expanding its squared norm separates individual step magnitudes from interactions between steps:

$$\begin{aligned} \|\Delta_T\|^2 = \eta^2 &\sum_{t=1}^{T} \|g_t\|^2 \\ &+ 2\eta^2 \sum_{1 \le u < t \le T} g_u^\top g_t. \end{aligned} \tag{7}$$

The terms $\|g_t\|^2$ are nonnegative and determine the squared lengths of individual SGD steps through $\eta^2\|g_t\|^2$. In full-batch GD, a larger gradient norm indicates a steeper local loss and, therefore, a larger first-order decrease along the negative gradient. The cross-step terms $g_u^\top g_t$ describe how these steps interact – positive inner products reinforce the displacement, whereas negative inner products cancel part of it. This decomposition provides the geometric basis for interpreting reinforcement and cancellation along a training trajectory.

To isolate directional alignment from gradient magnitudes, define the cosine similarity between any two nonzero gradients and its consecutive-step form

$$\begin{aligned} \rho_{u,t} &:= \frac{g_u^\top g_t}{\|g_u\|\|g_t\|}, \quad 1 \le u < t \le T, \\ \rho_t &:= \rho_{t-1,t}, \qquad\quad t = 2, \dots, T. \end{aligned} \tag{8}$$

We call the adjacent-gradient regime *constructive* when $\rho_t > 0$ and *destructive* when $\rho_t < 0$. Zero denotes orthogonal gradients; a zero gradient norm leaves its cosine undefined. Persistent signs identify sustained regimes, while sign changes mark transitions between them. These definitions refer to raw gradients throughout the paper; under plain SGD, their signs also describe the interaction between consecutive updates.

Substituting Eq. (8) into Eq. (7) gives

$$\begin{aligned} \|\Delta_T\|^2 = \eta^2 &\sum_{t=1}^{T} \|g_t\|^2 \\ &+ 2\eta^2 \sum_{1 \le u < t \le T} \|g_u\|\|g_t\|\rho_{u,t}. \end{aligned} \tag{9}$$

The exact sum retains interactions across every temporal lag. For given gradient magnitudes, positive pairwise terms reinforce net motion and negative terms cancel it. The adjacent statistic ρ_t makes the current regime observable online; longer-lag interactions determine how that local behavior contributes to motion over the full training horizon.

A. Directional Curvature and Consecutive Alignment

For mini-batch SGD, we explicitly incorporate stochastic gradient noise in Eq. (4), $g_t = \nabla\mathcal{L}(\theta_{t-1}) + \xi_t$. Suppose the full objective $\mathcal{L}$ is twice continuously differentiable along the preceding step, and write $H_{t-1} := \nabla^2\mathcal{L}(\theta_{t-2})$. Expanding the full gradient around θ_{t-2} and substituting the preceding SGD update yields

$$g_t = (I - \eta H_{t-1})g_{t-1} + (\xi_t - \xi_{t-1}) + r_t, \tag{10}$$

where r_t is the Taylor remainder. Neglecting r_t recovers the first-order approximation. Taking the inner product with g_{t-1} then gives

$$\begin{aligned} \|g_{t-1}\|\|g_t\|\rho_t \approx\ & \underbrace{\|g_{t-1}\|^2}_{\text{Gradient magnitude}} \\ & \underbrace{- \eta g_{t-1}^\top H_{t-1} g_{t-1}}_{\text{Curvature contribution}} \\ & \underbrace{+ g_{t-1}^\top(\xi_t - \xi_{t-1})}_{\text{Noise contribution}}. \end{aligned} \tag{11}$$

Positive curvature reduces the component of the next gradient along the preceding direction. Its strength relative to the preceding squared norm is the directional curvature

$$\kappa_t := \frac{g_{t-1}^\top H_{t-1} g_{t-1}}{\|g_{t-1}\|^2}, \tag{12}$$

Thus, κ_t measures how strongly a step changes the gradient along the direction actually followed, whereas ρ_t measures the resulting alignment. Sampling modifies this response through the normalized noise interaction

$$\varepsilon_t := \frac{g_{t-1}^\top(\xi_t - \xi_{t-1})}{\|g_{t-1}\|^2}. \tag{13}$$

Dividing Eq. (11) by the two gradient norms gives

$$\rho_t \approx \frac{\|g_{t-1}\|}{\|g_t\|}(1 - \eta\kappa_t + \varepsilon_t). \tag{14}$$

In full-batch GD, $\varepsilon_t = 0$. Within this first-order approximation, $\eta\kappa_t < 1$ retains a positive projection, while $\eta\kappa_t > 1$ reverses it: the next update partially opposes the preceding step. At $\eta\kappa_t = 1$, the new gradient has lost its component along the old direction. It is orthogonal if nonzero; if it vanishes, cosine is undefined. The norm ratio determines how this projected change translates into the magnitude of ρ_t.

The threshold at one also has an optimization interpretation. Along the full-gradient direction, the local quadratic loss model is $\mathcal{L}(\theta_{t-2} - \alpha g_{t-1}) \approx \mathcal{L}(\theta_{t-2}) - \alpha\|g_{t-1}\|^2 + \alpha^2\kappa_t\|g_{t-1}\|^2/2$. For $\kappa_t > 0$, its directional minimum is at $\alpha = 1/\kappa_t$. Negative alignment therefore identifies a step beyond this directional minimum under the approximation. This is the geometric meaning of the destructive regime: the next gradient opposes a component of the preceding step. The loss-increase boundary depends additionally on the strength of that opposition, as derived in Section III-D.

This interpretation leads to a measurable diagnostic. Rearranging Eq. (14) gives

$$\begin{aligned}\eta\widehat{\kappa}_t &:= 1 - \rho_t \frac{\|g_t\|}{\|g_{t-1}\|} \\ &= -\frac{g_{t-1}^\top (g_t - g_{t-1})}{\|g_{t-1}\|^2} \approx \eta\kappa_t - \varepsilon_t.\end{aligned} \tag{15}$$

This diagnostic measures the change of gradient projected onto its preceding direction, normalized by the preceding squared norm. It combines the regime with the relative gradient magnitude in one observable quantity. In full GD, a small Taylor remainder gives a directional-curvature estimate; under SGD, the explicit noise term describes the sampling contribution. The diagnostic concerns the direction actually traversed, complementing the largest-eigenvalue criterion in Eq. (3).

B. Noise Contribution under Mini-Batch Sampling

Sampling contributes to the regime through its interaction with the preceding noisy gradient. Write $g_{t-1} = \nabla\mathcal{L}(\theta_{t-2}) + \xi_{t-1}$. Fresh, conditionally unbiased batches give $\mathbb{E}[g_{t-1}^\top \xi_t] = 0$ and $\mathbb{E}[\nabla\mathcal{L}(\theta_{t-2})^\top \xi_{t-1}] = 0$. With finite second moments, the preceding gradient contributes its own noise energy:

$$\mathbb{E}[g_{t-1}^\top(\xi_t - \xi_{t-1})] = -\mathbb{E}[\|\xi_{t-1}\|^2] \le 0. \tag{16}$$

Thus, even unbiased sampling noise contributes a systematic nonpositive term to the expected unnormalized interaction between consecutive gradients.

This contribution is directly related to mini-batch size. Consider a mini-batch of b examples sampled uniformly without replacement from $N > 1$ training examples. Using the per-sample gradient covariance $\Sigma(\theta)$ defined in Eq. (5), the sampling variance satisfies

$$\mathbb{E}[\|\xi_{t-1}\|^2] = \frac{N-b}{b(N-1)}\mathbb{E}[\operatorname{tr}\Sigma(\theta_{t-2})]. \tag{17}$$

At fixed parameters, smaller batches produce larger sampling variance and, consequently, a larger negative expected noise contribution. For full-batch gradients ($b = N$), this contribution vanishes.

Noise can weaken directional coherence by enlarging gradient norms relative to their shared signal. For independent estimates at two fixed states, let μ, ν be the nonzero full gradients and v_μ, v_ν the expected squared noise norms. When sampled norms concentrate,

$$\mathbb{E}[\cos(\mu + \xi, \nu + \zeta)] \approx \frac{\cos(\mu, \nu)}{\sqrt{(1 + v_\mu/\|\mu\|^2)(1 + v_\nu/\|\nu\|^2)}}.$$

This attenuation moves either sign toward zero. At the same fixed state, the common full-gradient direction gives the special case

$$\mathbb{E}[\rho_t] \approx \frac{\|\nabla\mathcal{L}(\theta)\|^2}{\|\nabla\mathcal{L}(\theta)\|^2 + \frac{N-b}{b(N-1)}\operatorname{tr}\Sigma(\theta)}. \tag{18}$$

Here the expected cosine is nonnegative; the approximation quantifies the dilution of a common direction by sampling. During SGD, a sampled update also changes the next state, coupling noise and curvature. The fixed-state attenuation argument therefore isolates a mechanism, while the batch-size intervention in Section IV-D tests its expression along evolving trajectories.

C. Normalized Displacement and Cumulative Interference

The regime describes adjacent directions; normalized displacement describes their cumulative outcome. It compares the straight-line distance from initialization with the total distance traveled:

$$R_T := \frac{\|\Delta_T\|}{\eta\sum_{t=1}^T \|g_t\|} = \frac{\|\sum_{t=1}^T g_t\|}{\sum_{t=1}^T \|g_t\|}, \qquad 0 \le R_T \le 1, \tag{19}$$

For a nonzero accumulated path length under constant-step plain SGD, the denominator is the total distance traveled in parameter space, while the numerator measures the net displacement from initialization. Thus, R_T quantifies the fraction of accumulated step length retained as net displacement.

Using Eq. (9), this ratio satisfies

$$R_T^2 = \frac{\sum_{t=1}^T \|g_t\|^2 + 2\sum_{1\le u<t\le T}\|g_u\|\|g_t\|\rho_{u,t}}{\left(\sum_{t=1}^T \|g_t\|\right)^2}. \tag{20}$$

Thus R_T incorporates interactions across all temporal lags. When R_T is near one, almost all traveled distance survives as net displacement; when it is near zero, little does. Repeated back-and-forth steps and weakly coherent sampled directions both suggest low retention. For example, equal-length orthogonal updates give $R_T = 1/\sqrt{T}$, whereas equal opposing pairs cancel exactly. We therefore expect small R_T when such behavior persists over a substantial part of an SGD path. Absolute displacement can keep growing while its retained fraction falls. Together, ρ_t, gradient norms and R_T describe the local regime, the size of its steps and the fraction retained over time.

These diagnostics are inexpensive in computational structure. For p parameters, a norm and an inner product with the retained preceding gradient give ρ_t and $\eta\widehat{\kappa}_t$ in $O(p)$ operations and $O(p)$ additional storage per update. Computing R_T adds a running vector sum and a scalar sum of norms, also with

$O(p)$ work and storage independent of T. All use gradients already produced by the training backward pass; the exact all-pair identity is evaluated through these sums. Explicit lag profiles are a separate measurement requiring retained history.

D. From Destructive Descent to Loss Increase

A destructive step can still decrease loss after crossing its directional minimum. The decisive quantity is how strongly the new gradient opposes the preceding one. Define the full-objective change of that step as $\delta\mathcal{L}_{t-1} := \mathcal{L}(\theta_{t-1}) - \mathcal{L}(\theta_{t-2})$. The second-order approximation derived in Appendix A separates three contributions:

$$\begin{aligned}\delta\mathcal{L}_{t-1} \approx\ & \underbrace{-\frac{\eta}{2}\|g_{t-1}\|^2}_{\text{Gradient magnitude}} \\ & \underbrace{-\frac{\eta}{2}\rho_t\|g_{t-1}\|\|g_t\|}_{\text{Directional interference}} \\ & + \underbrace{\frac{\eta}{2}g_{t-1}^\top(\xi_{t-1}+\xi_t)}_{\text{Noise interaction}}.\end{aligned} \tag{21}$$

The magnitude term favors descent. The interference term reinforces it for positive alignment and opposes it for negative alignment: destructive interference contributes positively to the predicted loss change. With negligible noise and nonzero gradients, the approximation predicts a loss increase precisely when

$$\rho_t < -\frac{\|g_{t-1}\|}{\|g_t\|}. \tag{22}$$

The condition requires a destructive regime and sufficient gradient-norm growth: the opposing component of the new gradient must exceed the preceding gradient magnitude. This is the distinction that makes the joint diagnostic useful. The sign identifies the regime, while the norm ratio determines whether its destructive contribution can overwhelm the descent term.

Substituting Eq. (14) with zero noise gives

$$\delta\mathcal{L}_{t-1} \approx \frac{\eta}{2}\|g_{t-1}\|^2(\eta\kappa_t - 2). \tag{23}$$

Within the local quadratic model, $1 < \eta\kappa_t < 2$ is a destructive regime with decreasing loss; $\eta\kappa_t > 2$ produces a loss increase. The projected-change diagnostic expresses the same transition through observable gradient reversal and amplification. Nonlinear remainder and SGD noise terms delimit this approximation. We now test how the predicted regimes appear in neural-network training and how their signatures relate to cumulative motion and loss spikes.

IV. Constructive and Destructive Regimes in GD and SGD

We first examine whether destructive regimes sustain descent and how their reversals shape cumulative parameter motion. Full-batch GD isolates these dynamics on fixed objectives; matched mini-batch continuations then show how sampling changes the regimes encountered during training.

The comparison covers a tanh MLP on 5,000 CIFAR-10 images, ResNet-32 on all 50,000 CIFAR-10 training images, and a word-level Transformer on WikiText-2. Each minimizes its complete fixed training objective for 3,000 updates at three constant learning rates (Table I). Within each model, rates share one frozen initialization and data preparation; they were selected by earlier training calibrations to expose contrasting dynamics, without test-loss selection. Plain GD uses no momentum, weight decay, clipping or schedule. The three cases test whether the same geometric signatures recur across different objectives and architectures. Appendix C specifies objectives, architectures, precision and measurement details.

A. Descent in a Destructive Regime

Figure 1 pairs loss with the projected gradient change $\eta\widehat{\kappa}_t$. The higher-rate trajectories develop oscillatory loss while continuing to improve over longer timescales; their late projected change approaches two. Their negative adjacent cosines in Fig. 2 identify a sustained destructive regime. Equation (23) explains why this value is informative: in the local quadratic model, it separates a loss-decreasing step from one that reverses the preceding progress. The observed pattern gives a gradient-based view of the oscillatory descent associated with the Edge of Stability [1].

Consecutive gradients therefore expose an oscillatory regime that is productive over the training horizon. The projected-change diagnostic provides its directional interpretation using inexpensive vector reductions. Appendix C evaluates the finite-step loss approximation and supplies complementary spectral measurements with their objective coverage.

B. Gradient Magnitude and Cumulative Motion

Figure 2 explains how the destructive regime changes parameter motion. The lowest MLP and Transformer rates remain constructive, whereas the other seven conditions develop nearly opposing adjacent gradients. In the MLP and ResNet, this transition also accompanies growing gradient norms: the steps become larger even as their directions increasingly cancel. These fluctuations arise without mini-batch sampling.

The decreasing R_t quantifies the cancellation associated with these reversals. A growing fraction of the traveled distance is spent moving back and forth, even as the parameters move farther from initialization. Table I shows the resulting combination: within every model, the highest tested rate has the lowest retention, the largest absolute displacement, and the lowest final training loss. A rate can therefore be large relative to local directional minima yet productive over the full training horizon.

Local destruction coexists with reinforcement over longer horizons. The complete cross-step interaction is positive at the endpoint in all nine trajectories, including the seven with negative accumulated lag-one interaction (Table III). Longer-range reinforcement can coexist with repeated local reversal. Together, ρ_t, gradient norms and R_t distinguish local overshoot, step size and retained motion. Their joint interpretation explains how oscillatory training continues to make objective progress.

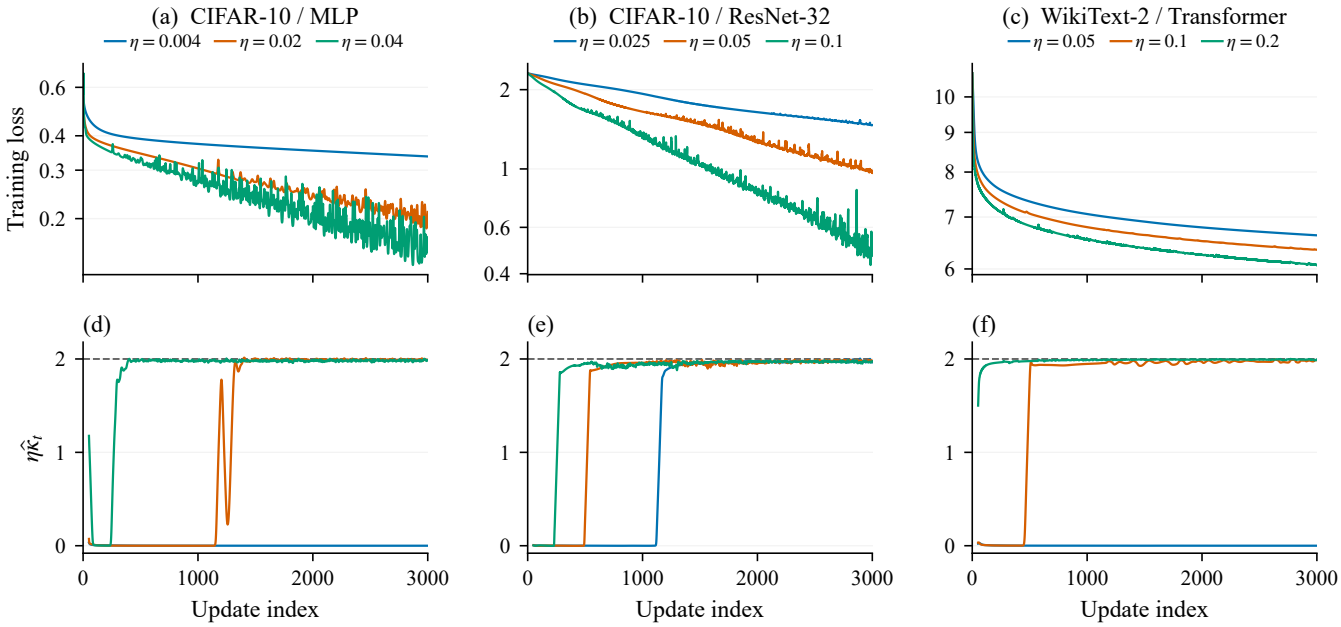


Fig. 1: Training loss and projected gradient change under full-batch GD. Columns identify the dataset/model; colors identify learning rates within each column. Top: unsmoothed full-objective loss on logarithmic axes. Bottom: complete 50-pair trailing means of the pointwise $\eta\widehat{\kappa}_t$, with the local quadratic loss-change boundary at two. Loss at index t is evaluated at θ_t; gradient diagnostics use g_t at θ_{t-1}. Each model uses one shared initialization. Trailing means summarize sustained regimes.

TABLE I: Endpoints at $T = 3000$ updates. The late means $\overline{\rho}_{\text{late}}$ and $\overline{\eta\widehat{\kappa}}_{\text{late}}$ average gradient indices $t = 2001, \ldots, 3000$. Displacement is $\|\Delta_T\| = \eta\|\sum_{t=1}^{T} g_t\|$, subject to parameter-update roundoff. Each model shares one initialization across rates; comparisons are within models. *Positive value below 0.00005, rounded to zero.

Dataset	Model	η	$\mathcal{L}(\theta_T)$	$\overline{\rho}_{\text{late}}$	R_T	$\overline{\eta\widehat{\kappa}}_{\text{late}}$	$\|\Delta_T\|$
CIFAR-10	MLP	0.004	0.3365	1.0000	0.7941	0.0000*	0.9511
		0.02	0.2114	-0.9913	0.0488	1.9924	3.1391
		0.04	0.1539	-0.9799	0.0257	1.9820	4.7628
	ResNet-32	0.025	1.4665	-0.9723	0.1385	1.9740	5.7045
		0.05	0.9672	-0.9708	0.0616	1.9746	9.0183
		0.1	0.4898	-0.9684	0.0374	1.9707	13.9329
WikiText-2	Transformer	0.05	6.6301	1.0000	0.7223	0.0002	11.3183
		0.1	6.3519	-0.9749	0.1395	1.9754	15.6026
		0.2	6.0697	-0.9937	0.0541	1.9937	21.9693

C. Oscillation Beyond Adjacent Updates

To examine the temporal extent of reversal, we fix one endpoint gradient and compare it with preceding gradients at every lag $k = 1, \ldots, 256$. Figure 3 uses the largest tested rate for each model and the common endpoint after 2,256 updates. Strongly opposing and reinforcing pairs recur at longer separations, including after intermediate lags with near-zero similarity. The varying amplitude and sign pattern show that oscillation extends beyond a single adjacent pair, while its persistence depends on the model and part of the trajectory. Adjacent cosine identifies the local regime within this longer temporal structure; cumulative retention summarizes its net geometric effect.

D. How Sampling Changes the Regime

We vary batch size at initialization and after 2,000 GD updates in the same three model–dataset cases, holding $\eta = 0.02$, 0.1 and 0.2 fixed for the MLP, ResNet-32 and Transformer, respectively. Each starting state receives 200-update continuations, with three initializations, eight sampling repetitions per stochastic condition and one full-GD control per initialization. Figure 4 compares mean sampled consecutive-gradient cosine over each continuation. Equal update budgets imply unequal data exposure; sampling units, objective coverage and numerical checks are in Appendix C.

The smallest batches weaken the trained-state destructive regime in all three cases. At initialization, full GD gives a constructive mean for MLP and ResNet, while the Trans-

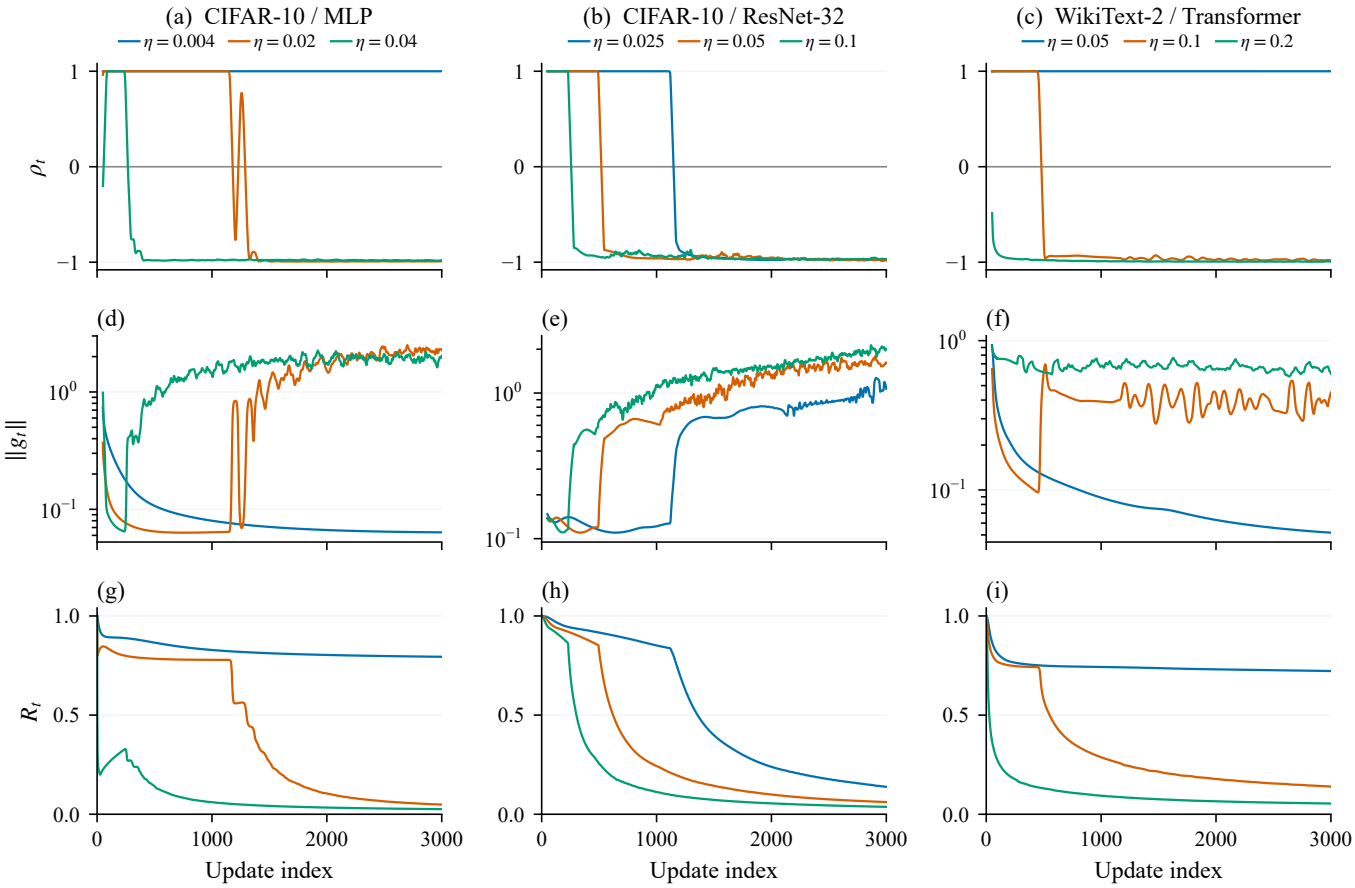


Fig. 2: Directional coherence and cumulative motion for the conditions in Fig. 1. Rows show consecutive cosine, gradient norm (logarithmic scale), and normalized displacement. Cosines and norms use complete 50-observation trailing means; R_t is the unsmoothed ratio over the complete prefix through update t. Colors and column ordering match Fig. 1. Raw cosine traces appear in Appendix C.

former develops a strongly destructive regime within the first 200 updates. The MLP's sampled paths can instead develop negative mean alignment. Sampling therefore influences the regime by changing both the observed gradients and the states that subsequent updates visit.

For the Transformer at $b = 8$, both starting states have mean cosine near -0.26, yet mean full-data loss falls from initialization and rises after GD training. The MLP's smallest trained-state batch likewise raises mean full-data loss, whereas full GD lowers it in both models. These outcomes reinforce the role of the joint diagnostics: cosine identifies the directional regime, and gradient magnitudes and loss reveal its consequences. The next section examines this distinction under Adam, where temporal averaging and adaptive scaling further transform the gradient before each step.

V. Diagnosing Loss Spikes under Adam

Adam provides a practical test of the central argument: can raw-gradient regimes reveal developing loss instability when the optimizer takes an adaptive step? The diagnostics in Eqs. (8) and (15) measure how the loss gradient changes between successive parameter states. That information remains relevant along Adam's path because the raw gradients both describe the encountered landscape and supply the input to its adaptive update.

Adam [11] averages this input through first and second moments,

$$\begin{aligned} m_t &= \beta_1 m_{t-1} + (1-\beta_1) g_t, \\ v_t &= \beta_2 v_{t-1} + (1-\beta_2) g_t^{\odot 2}, \end{aligned} \tag{24}$$

starting from zero. Bias correction gives $\widehat{m}_t = m_t/(1-\beta_1^t)$ and $\widehat{v}_t = v_t/(1-\beta_2^t)$, followed by

$$d_t = \frac{\widehat{m}_t}{\sqrt{\widehat{v}_t} + \epsilon}, \qquad \theta_t = \theta_{t-1} - \eta d_t, \tag{25}$$

with coordinatewise operations. The first moment carries information from preceding gradients, while the second moment scales coordinates. A sharp change in the raw gradient therefore enters an optimizer whose response also depends on its accumulated history.

The connection to the loss landscape follows from the same local expansion used for GD. For full-batch gradients along

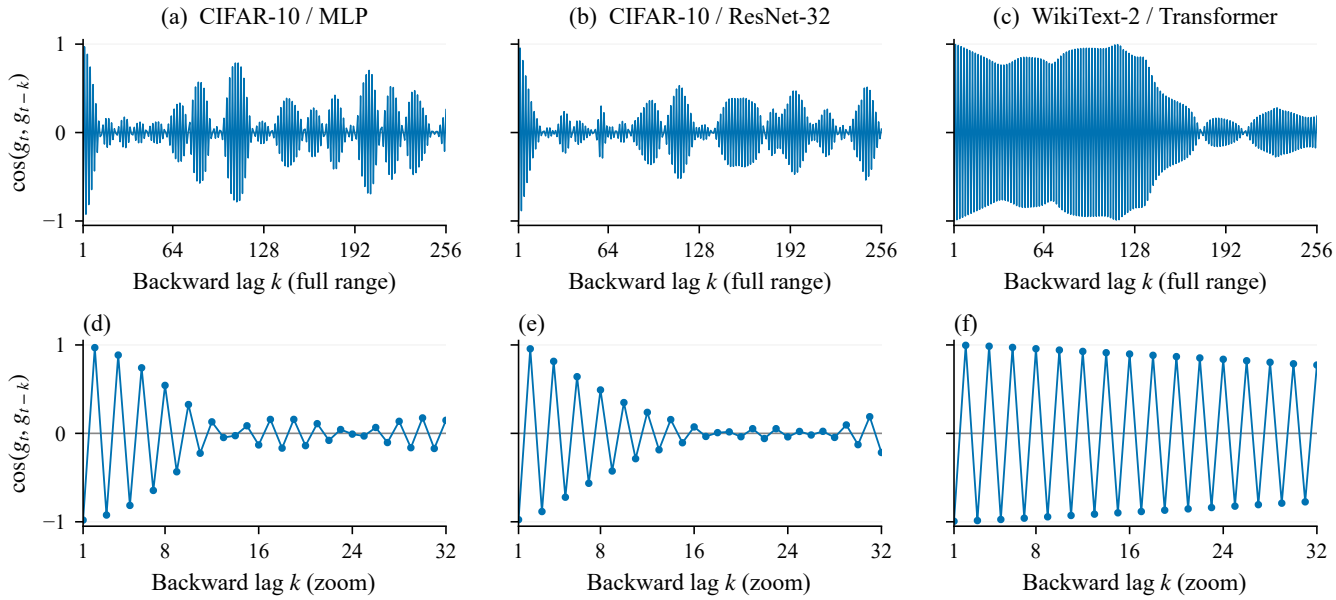


Fig. 3: Backward gradient alignment at the common endpoint state θ_{2256}, corresponding to $t = 2257$ under the manuscript's preupdate gradient indexing. Each point is one exact cosine $\cos(g_t, g_{t-k})$; the endpoint is fixed across all 256 lags. The top row shows lags 1–256; the bottom row magnifies the same values at lags 1–32 to resolve the early sign alternation. Rates are 0.04 (MLP), 0.1 (ResNet-32) and 0.2 (Transformer). These are separately recomputed 256-update GD continuations from retained state 2000, with one initialization and no smoothing or endpoint averaging. Appendix C records the additional endpoints and numerical replay scope.

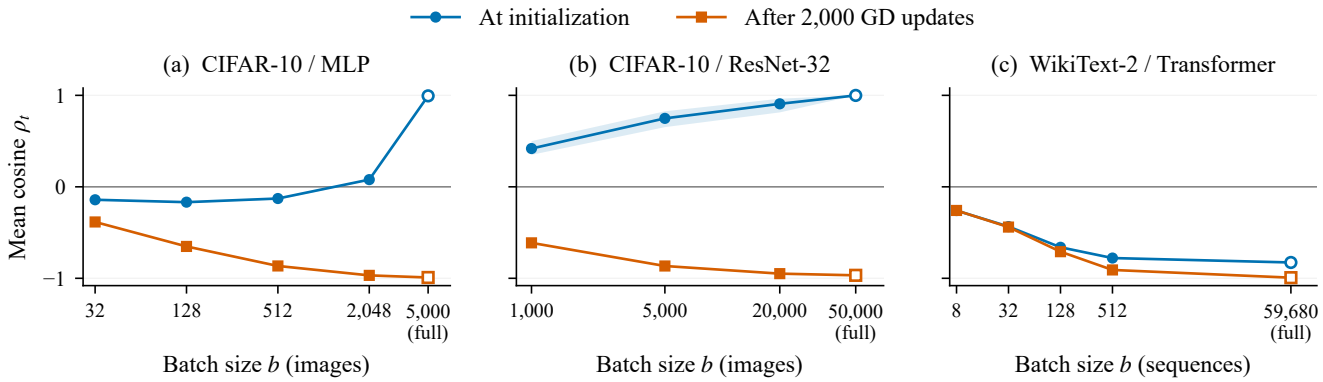


Fig. 4: Mean sampled-gradient cosine over 199 adjacent pairs during 200-update continuations in three model–dataset cases. Blue circles start at initialization; orange squares start after 2,000 GD updates. Each line averages repetitions within initialization, then weights the three initialization blocks equally; shading is their range, not a confidence interval. Hollow rightmost markers denote full GD. ResNet batches contain whole fixed 1,000-image normalization groups; Transformer batches count prepared sequences. Batch axes are logarithmic; lines guide the eye between measured sizes.

Adam's step,

$$\eta\widehat{\kappa}_t = -\frac{g_{t-1}^\top (g_t - g_{t-1})}{\|g_{t-1}\|^2} \approx \eta\,\frac{g_{t-1}^\top H_{t-1} d_{t-1}}{\|g_{t-1}\|^2}. \tag{26}$$

Thus the diagnostic measures the gradient's projected response to the adaptive displacement. Its exact expression $1 - \rho_t\|g_t\|/\|g_{t-1}\|$ separates that response into the current regime and relative gradient magnitude. In a destructive regime, an increase in the opposing gradient component raises $\eta\widehat{\kappa}_t$; exceeding two means that component is larger than the preceding gradient. The formal GD loss boundary motivates this reference, while the Adam experiments below establish its empirical relation to developing spikes.

A. Controlled Training and Loss Overview

We use the CIFAR-10/MLP, CIFAR-10/ResNet-32 and WikiText-2/Transformer objectives from Section IV. Each trajectory starts from the same frozen, untrained seed-0

weights as its GD counterpart, with fresh Adam moments. Each model receives 1,024 full-objective updates at three constant rates: $\{10^{-4}, 3\times10^{-4}, 10^{-3}\}$ for the MLP and $\{3\times10^{-4}, 10^{-3}, 3\times10^{-3}\}$ for ResNet and Transformer. We use $(\beta_1,\beta_2) = (0.9, 0.999)$ and $\epsilon = 10^{-8}$. Full-batch accumulation, fixed rates and the absence of clipping or weight decay isolate the diagnostic patterns within the adaptive training dynamics. Appendix C-G gives the numerical and measurement details.

Figure 5 shows sustained loss reduction interrupted by transient spikes, especially at larger rates. The highest tested rate attains the lowest final training loss in every model. The practical problem is therefore to understand and potentially attenuate the excursions within an otherwise productive trajectory. We begin with individual spikes to resolve their temporal structure, then examine their recurrence across all nine runs.

B. Early Signs and Anatomy of Individual Spikes

For each model, we select the largest recovered relative loss spike at its highest tested learning rate. This loss-based rule selects peaks at states 703, 422 and 794 for MLP, ResNet and Transformer, respectively. Figure 6 shows their raw-gradient and momentum dynamics at individual-update resolution.

All three spikes develop from a destructive raw-gradient regime. As the loss rises, gradient norms grow sharply and $\eta\widehat{\kappa}$ exceeds two before the peak. The momentum norm also grows, showing that the amplified gradients enter Adam's temporal average. The joint view explains the diagnostic: negative ρ identifies the reversal, and the growing norm makes that reversal increasingly strong.

The Transformer gives a particularly clear early warning. At state 788, $\eta\widehat{\kappa}$ first exceeds two while the loss is only 0.40% above its pre-rise level. Two updates later the increase exceeds 10%; the loss reaches its maximum six updates after the diagnostic crossing, at state 794. Gradient and momentum norms peak at states 792 and 793. The diagnostic exposes the developing excursion before the pronounced rise, providing a concrete interval in which an intervention could act.

Recovery reveals how the regime changes after amplification. In the displayed ResNet and Transformer events, the gradients enter constructive segments as projected change falls and loss declines smoothly. The MLP retains a predominantly destructive raw-gradient regime and alternating momentum alignment, with repeated rebounds during recovery. Both patterns reduce the spike, but through different directional dynamics: constructive descent in the former examples and shrinking oscillation in the MLP.

These cases connect a visible loss spike to an earlier change in the gradient field traversed by Adam. Their shared feature is amplification within a destructive regime; their different recovery paths show why monitoring the regime and magnitude together is informative throughout the event. We next test how broadly this anatomy recurs.

C. Defining and Measuring Recurrent Spikes

A spike is an excursion above a fixed pre-rise loss: the loss leaves that level, reaches a maximum, and eventually returns. Keeping the reference fixed treats a rise with several local maxima as one event until recovery. This gives each event a clear beginning, peak and duration using the loss alone.

Write $\mathcal{L}_s = \mathcal{L}(\theta_s)$ for the loss after s updates. For event i, let a_i denote the state immediately before its rise, p_i its highest-loss state, and c_i its first return to the pre-rise level. Starting from $c_0 = 0$, define

$$\begin{aligned} a_i &:= \min\{s : c_{i-1} \le s < T,\ \mathcal{L}_{s+1} > \mathcal{L}_s\},\\ c_i &:= \min\{s : a_i < s \le T,\ \mathcal{L}_s \le \mathcal{L}_{a_i}\},\\ p_i &:= \min \arg\max_{a_i<s<c_i} \mathcal{L}_s. \end{aligned} \tag{27}$$

The first elevated loss occurs at $a_i + 1$, and the earliest maximum defines p_i if there is a tie. We retain events with observed recovery by $T = 1024$ and resume the search at c_i; the search ends when a rise or a subsequent return is absent. All measured pre-rise losses are positive. The peak increase is $100(\mathcal{L}_{p_i}/\mathcal{L}_{a_i} - 1)$ percent; rise and recovery times are $p_i - a_i$ and $c_i - p_i$ updates. Relative to the same baseline, the loss ratios before, at the peak and at recovery are 1, $\mathcal{L}_{p_i}/\mathcal{L}_{a_i}$ and $\mathcal{L}_{c_i}/\mathcal{L}_{a_i} \le 1$.

The distributions in Fig. 7 show that the selected cases belong to a recurring pattern. Larger learning rates extend the distribution toward more severe excursions, while small fluctuations remain common. We focus the anatomy analysis on increases strictly greater than 10%, an amplitude criterion that selects pronounced departures from the local training loss. This yields 129 recovered spikes; the full density places their amplitudes in context.

Table II exposes the cost of these excursions. Both mean and maximum spike increase grow with learning rate among the eligible rates, reaching a maximum of 859% in ResNet. Recovery is substantially slower than the rise in ResNet (17–19 versus 6–8 updates) and Transformer (24–31 versus about 6). A short amplification therefore interrupts progress for a much longer interval. MLP rise and recovery times are more comparable, consistent with its repeated oscillatory pattern.

The regime and norm statistics identify the recurring anatomy. Pre-rise ρ is negative in 117 of 127 defined events, and gradient norms increase from pre-rise to the loss peak in all 129. The diagnostic $\eta\widehat{\kappa}$ is already above two before the rise in 21 of 127 defined cases; its event maximum precedes the loss peak in 82 of 129 events. Thus the most consistent population signature is a destructive initial regime followed by gradient amplification, with projected change often responding ahead of peak loss. During recovery, mean gradient norm ratios fall below one in every group. The MLP and ResNet retain negative group-mean recovery cosines, while Transformer means approach zero: the population also exhibits recovery through shrinking destructive oscillations.

D. Regime, Amplification and Spike Magnitude

The preceding results suggest that spike severity depends jointly on the regime before the rise and the gradient amplification during it. We summarize this combination by plotting

$$\rho_{a_i+1}\frac{\|g_{p_i+1}\|}{\|g_{a_i+1}\|} \quad \text{against} \quad \frac{\mathcal{L}_{p_i}}{\mathcal{L}_{a_i}}. \tag{28}$$

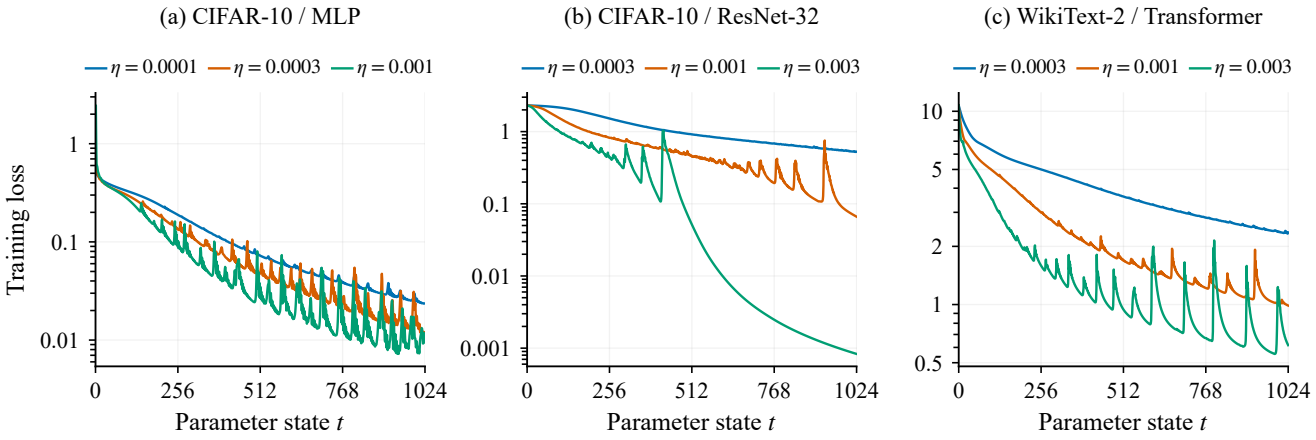


Fig. 5: Training loss under full-batch Adam at three constant learning rates per model. Rates share one frozen initialization within each model. Every loss observation is shown without smoothing on logarithmic y-axes. Parameter state s denotes s completed updates, from initialization through update 1,024. Colors identify the rates in each panel.

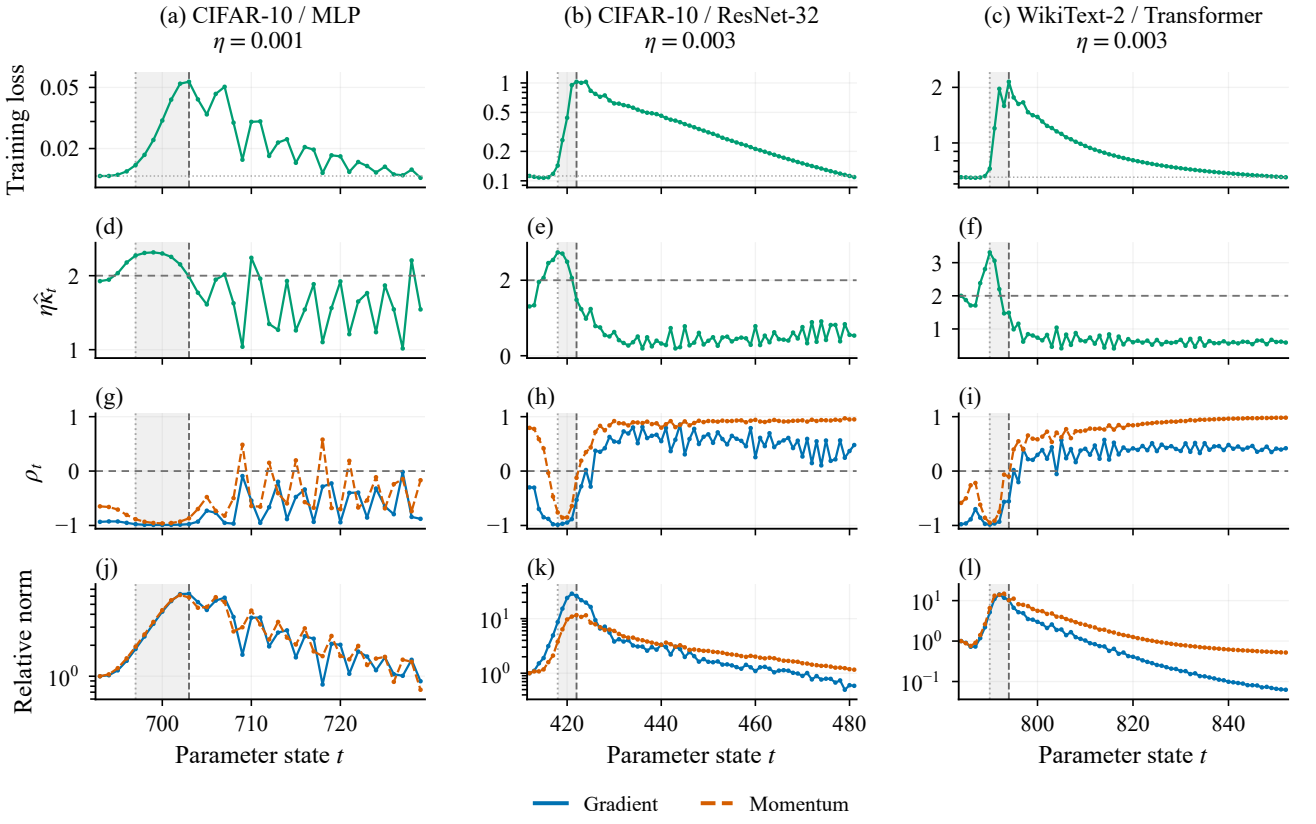


Fig. 6: Largest recovered relative loss spike at each model's highest Adam rate. Rows show training loss, raw-gradient projected change, adjacent cosines of raw gradients (blue) and first moments (orange), and their relative norms. All values are unsmoothed. At parameter state s, these diagnostics use g_{s+1} and m_{s+1}. Gray bands mark the pronounced rise from the first +10% crossing to the loss peak: 697–703, 418–422 and 790–794. Norms are normalized by their first displayed value. The dotted loss reference is the first displayed loss; windows begin ten states before the peak and end at the first later loss below that reference.

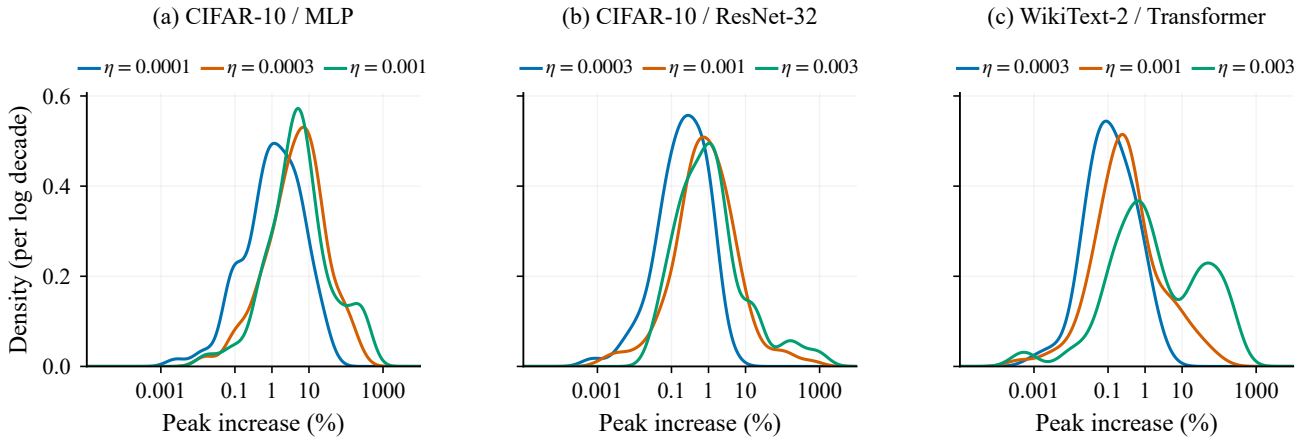


Fig. 7: Peak-increase distributions for all 806 recovered loss excursions. Colors identify learning rates within each model. Gaussian kernel densities use $\log_{10}$ percentage-increase coordinates and a common pooled bandwidth within each model. Each rate's density integrates to one in that coordinate. The logarithmic x-axis shows percentage increase: 1 denotes +1% and 100 denotes +100%.

TABLE II: Anatomy of recovered loss spikes with peak increase $> 10\%$. η is the constant Adam learning rate and N the number of events. States a_i, p_i and c_i mark pre-rise, peak and first recovery to the pre-rise loss. Entries are event means $\pm$ sample SD, except N and the maximum. Peak increase is $100(\mathcal{L}_{p_i}/\mathcal{L}_{a_i} - 1)$; rise and recovery times are $p_i - a_i$ and $c_i - p_i$ updates. Raw denotes gradients of the training objective before Adam's transformations: $\rho_t = \cos(g_{t-1}, g_t)$ and $\eta\widehat{\kappa}_t = 1 - \rho_t\|g_t\|/\|g_{t-1}\|$. Before uses the single observation at θ_{a_i} (gradient index $a_i + 1$); recovery summarizes each event's mean over states $p_i+1, \ldots, c_i$. Gradient norm ratios are $\|g_{p_i+1}\|/\|g_{a_i+1}\|$ during the rise and $\|g_{c_i+1}\|/\|g_{p_i+1}\|$ during recovery. Dashes indicate no qualifying event. Two initialization events have undefined before diagnostics, giving 46 and 38 observations for those entries at the MLP's middle and highest rates.

Dataset Model	η	N	Peak increase (%)		Time (updates)		$\eta\widehat{\kappa}$ (raw)		ρ (raw)		Gradient norm ratio	
			Mean $\pm$ SD	Max	Rise	Recovery	Before	Recovery	Before	Recovery	Rise	Recovery
CIFAR-10 MLP	10^{-4}	8	19 ± 6.1	31	8.4 ± 2.9	4.4 ± 2.4	1.8 ± 0.38	1.7 ± 0.08	-0.77 ± 0.31	-0.89 ± 0.08	5.3 ± 1.7	0.30 ± 0.07
	3×10^{-4}	47	43 ± 46	203	5.5 ± 4.4	4.9 ± 4.0	1.3 ± 0.25	1.6 ± 0.18	-0.53 ± 0.29	-0.71 ± 0.24	4.5 ± 3.0	0.33 ± 0.13
	10^{-3}	39	95 ± 107	318	6.1 ± 5.9	7.4 ± 7.5	1.4 ± 0.40	1.4 ± 0.26	-0.46 ± 0.38	-0.55 ± 0.30	5.2 ± 3.8	0.31 ± 0.14
CIFAR-10 ResNet-32	3×10^{-4}	0	–	–	–	–	–	–	–	–	–	–
	10^{-3}	10	105 ± 182	602	7.5 ± 2.2	17 ± 11	1.3 ± 0.39	1.3 ± 0.25	-0.36 ± 0.42	-0.36 ± 0.26	4.3 ± 2.9	0.43 ± 0.24
	3×10^{-3}	7	179 ± 309	859	6.3 ± 1.3	19 ± 20	1.7 ± 0.38	1.1 ± 0.37	-0.65 ± 0.28	-0.13 ± 0.42	5.0 ± 4.2	0.34 ± 0.24
WikiText-2 Transformer	3×10^{-4}	0	–	–	–	–	–	–	–	–	–	–
	10^{-3}	6	33 ± 25	81	6.3 ± 1.9	24 ± 9.8	2.1 ± 0.27	0.96 ± 0.16	-0.84 ± 0.10	-0.02 ± 0.12	9.0 ± 3.8	0.09 ± 0.05
	3×10^{-3}	12	82 ± 70	231	6.2 ± 2.7	31 ± 17	2.2 ± 0.43	0.91 ± 0.19	-0.79 ± 0.22	0.09 ± 0.21	11 ± 6.9	0.08 ± 0.08

The horizontal coordinate preserves the sign of the pre-rise regime and weights it by the rise norm ratio. More negative values combine stronger initial opposition with larger subsequent amplification. It is a retrospective event descriptor: Eq. (15) instead uses an adjacent norm ratio to describe the response at each update.

Figure 8 shows a clear association between this joint descriptor and spike magnitude. Separate fits at each learning rate explain 67–69% of variation in log relative peak loss for the MLP, 90–94% for ResNet, and 38–54% for the Transformer. The strongest relation occurs in ResNet, while the Transformer retains more variation beyond the two-factor description. These results extend the micro-plots: the destructive regime and gradient amplification jointly organize the severity of repeated spikes across the three cases.

This anatomy gives a concrete rationale for testing interventions during the developing rise. Gradient clipping directly controls the large input norms implicated by the analysis, consistent with its use against exploding gradients [19] and theoretical accounts of adaptivity under gradient-dependent smoothness [20]. For Adam, clipping changes both moment streams; learning-rate adjustment changes the adaptive step, and data mixing changes the gradients entering it. The observed early diagnostic response motivates applying these interventions before the pronounced loss increase and comparing them with unchanged continuations from the same saved state.

The present evidence establishes temporal warning signs and within-run associations in three controlled model–dataset

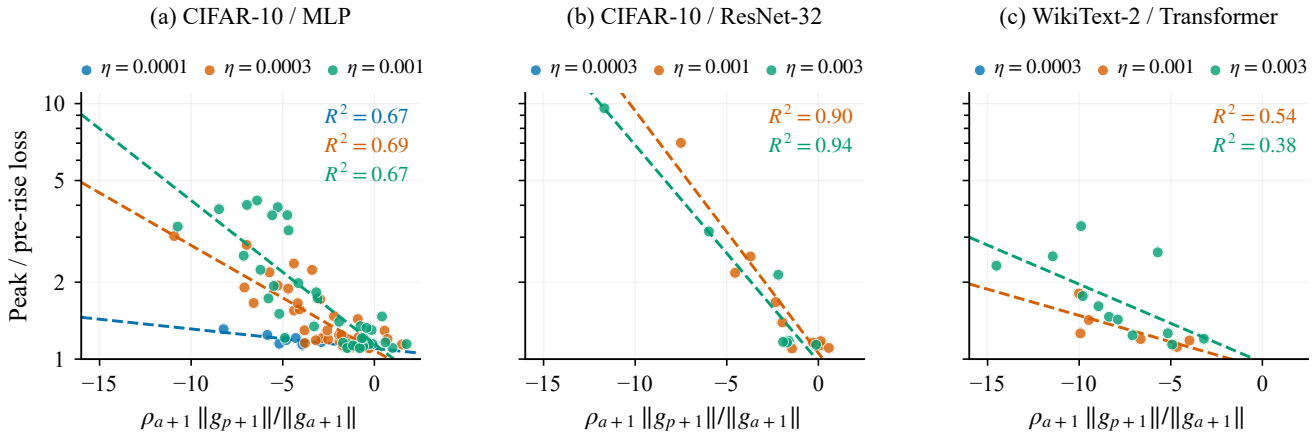


Fig. 8: Pre-rise raw-gradient cosine multiplied by the rise norm ratio versus relative peak loss, as defined in Eq. (28). Each point is one of 127 recovered spikes above 10% with defined pre-rise cosine. Colors identify learning rates. Dashed per-rate least-squares lines fit $\log_{10}$(peak/pre-rise loss) with an intercept and extend across the panels. Color-matched R^2 values measure explained variation on that log response in the fitted sample. The logarithmic y-axis shows a loss ratio: 1 denotes no increase and 2 denotes +100%.

cases. Events share trajectories, and the scatter includes gradient norms at the eventual loss peak. Prospective detection and causal tests are therefore the next stage: use only information available at the decision state, include non-spike periods, and measure both spike attenuation and subsequent loss at matched training budgets.

VI. Conclusion

Constructive and destructive regimes provide an interpretable view of neural-network training through gradients already computed by the optimizer. Adjacent cosine identifies the regime, gradient norms measure its amplification, and normalized displacement captures cumulative motion. The local loss relation explains the central distinction: a destructive regime can sustain descent, while sufficiently strong opposing gradients can produce loss-increasing steps. Controlled GD and SGD experiments show how these regimes evolve with learning rate and mini-batch sampling.

The Adam results establish the relevance of the same raw-gradient view along adaptive trajectories. Diagnostic changes precede the largest loss peaks, and the event analysis connects spike magnitude to a predominantly destructive initial regime followed by gradient amplification. The micro-plots supply temporal evidence, while the distributions and regressions show how the pattern repeats across models and rates. Together, they identify observable early signs and a consistent anatomy of local loss instability.

The resulting framework combines inexpensive measurement with a concrete interpretation of the trajectory. It motivates interventions timed to gradient amplification within a destructive regime, including learning-rate adjustment, gradient clipping and data mixing. Prospective warning rules and matched intervention tests are the next step toward using these diagnostics to reduce spikes while preserving training progress.

Appendix A
Derivation of the Alignment-Based Loss Change

The loss relation in Section III-D combines gradient magnitude and regime in a single expression. To derive it, consider the SGD update $\theta_{t-1} = \theta_{t-2} - \eta g_{t-1}$. The change in the full objective along this step can be expressed exactly as

$$\begin{aligned}&\mathcal{L}(\theta_{t-1}) - \mathcal{L}(\theta_{t-2})\\ &\quad = -\eta \int_0^1 g_{t-1}^\top \nabla\mathcal{L}(\theta_{t-2} - s\eta g_{t-1})\, ds.\end{aligned} \tag{29}$$

Approximating the integral by the average of the full gradients at the two endpoints gives

$$\begin{aligned}&\mathcal{L}(\theta_{t-1}) - \mathcal{L}(\theta_{t-2})\\ &\quad \approx -\frac{\eta}{2} g_{t-1}^\top \left(\nabla\mathcal{L}(\theta_{t-2}) + \nabla\mathcal{L}(\theta_{t-1})\right).\end{aligned} \tag{30}$$

This approximation is exact for quadratic objectives. For nonlinear objectives, the approximation error is of third order in the step length when the Hessian is locally Lipschitz continuous.

Using the stochastic gradient decomposition $g_t = \nabla\mathcal{L}(\theta_{t-1}) + \xi_t$ at both endpoints, we obtain

$$\begin{aligned}&\mathcal{L}(\theta_{t-1}) - \mathcal{L}(\theta_{t-2})\\ &\quad \approx -\frac{\eta}{2} g_{t-1}^\top (g_{t-1} + g_t - \xi_{t-1} - \xi_t)\\ &\quad = -\frac{\eta}{2}\left(\|g_{t-1}\|^2 + g_{t-1}^\top g_t\right)\\ &\qquad + \frac{\eta}{2} g_{t-1}^\top (\xi_{t-1} + \xi_t).\end{aligned} \tag{31}$$

Substituting $g_{t-1}^\top g_t = \|g_{t-1}\|\|g_t\|\rho_t$ and factoring the preceding squared gradient norm yields

$$\begin{aligned}&\mathcal{L}(\theta_{t-1}) - \mathcal{L}(\theta_{t-2})\\ &\quad \approx -\frac{\eta}{2}\|g_{t-1}\|^2 \left(1 + \rho_t \frac{\|g_t\|}{\|g_{t-1}\|}\right)\\ &\qquad + \frac{\eta}{2} g_{t-1}^\top (\xi_{t-1} + \xi_t),\end{aligned} \tag{32}$$

which recovers Eq. (21).

To connect this expression with directional curvature, recall from Eq. (14) that

$$\rho_t \frac{\|g_t\|}{\|g_{t-1}\|} \approx 1 - \eta\kappa_t + \varepsilon_t. \tag{33}$$

Substituting this relation and using the definition of ε_t in Eq. (13) gives

$$\begin{aligned}&\mathcal{L}(\theta_{t-1}) - \mathcal{L}(\theta_{t-2}) \\ &\quad\approx -\eta\|g_{t-1}\|^2\left(1 - \frac{\eta\kappa_t}{2}\right) + \eta\xi_{t-1}^\top g_{t-1}.\end{aligned} \tag{34}$$

Thus, the curvature-based and alignment-based expressions describe the same local loss change to second order. Positive directional curvature counteracts the first-order loss decrease. The weighted alignment expresses that effect through the regime and gradient-norm ratio: constructive interaction strengthens descent, while sufficiently amplified destructive interaction reverses it. The noise contributions account for the difference between the observed mini-batch gradients and the full gradients.

Appendix B
Derivation of the Local Loss Changes

We use the same indexing as Section III: g_{t-1} is evaluated at θ_{t-2}, g_t at θ_{t-1}, and $H_{t-1} = \nabla^2\mathcal{L}(\theta_{t-2})$. The loss is assumed twice continuously differentiable along the displacements considered below. The relations retain second-order terms and omit the Taylor remainders; they are exact when the objective is quadratic.

A. One Update and Directional Curvature

For $\theta_{t-1} - \theta_{t-2} = -\eta g_{t-1}$, Taylor expansion gives

$$\begin{aligned}&\mathcal{L}(\theta_{t-1}) - \mathcal{L}(\theta_{t-2}) \\ &\quad\approx -\eta\nabla\mathcal{L}(\theta_{t-2})^\top g_{t-1} + \frac{\eta^2}{2} g_{t-1}^\top H_{t-1} g_{t-1}.\end{aligned} \tag{35}$$

Substituting $\nabla\mathcal{L}(\theta_{t-2}) = g_{t-1} - \xi_{t-1}$ and $g_{t-1}^\top H_{t-1} g_{t-1} = \kappa_t\|g_{t-1}\|^2$ yields Eq. (34) for nonzero g_{t-1}. The projected-gradient relation in Eq. (15) then gives the alignment form in Eq. (21).

The first-order noise contribution combines with the noise contained in the squared gradient norm. Conditional on the parameters before the batch is drawn, unbiased sampling gives

$$\begin{aligned}\mathbb{E}[\|g_{t-1}\|^2 \mid \theta_{t-2}] &= \|\nabla\mathcal{L}(\theta_{t-2})\|^2 \\ &\quad + \mathbb{E}[\|\xi_{t-1}\|^2 \mid \theta_{t-2}], \\ \mathbb{E}[\xi_{t-1}^\top g_{t-1} \mid \theta_{t-2}] &= \mathbb{E}[\|\xi_{t-1}\|^2 \mid \theta_{t-2}].\end{aligned} \tag{36}$$

Thus, the contributions from noise variance cancel between the first-order terms when taking their conditional expectation. With finite moments for the retained terms, the expected loss change in the quadratic approximation is

$$\begin{aligned}&\mathbb{E}[\mathcal{L}(\theta_{t-1}) - \mathcal{L}(\theta_{t-2}) \mid \theta_{t-2}] \\ &\quad\approx -\eta\|\nabla\mathcal{L}(\theta_{t-2})\|^2 \\ &\qquad + \frac{\eta^2}{2}\mathbb{E}[g_{t-1}^\top H_{t-1} g_{t-1} \mid \theta_{t-2}].\end{aligned} \tag{37}$$

Sampling contributes through the curvature-weighted second-order term. For a nonlinear objective, the Taylor remainder supplies the additional finite-step correction.

B. Two Updates and Consecutive Alignment

The displacement over two SGD updates is exactly

$$\theta_t - \theta_{t-2} = -\eta(g_{t-1} + g_t).$$

Expanding around the starting parameters θ_{t-2} gives

$$\begin{aligned}&\mathcal{L}(\theta_t) - \mathcal{L}(\theta_{t-2}) \\ &\quad\approx -\eta\nabla\mathcal{L}(\theta_{t-2})^\top(g_{t-1} + g_t) \\ &\qquad + \frac{\eta^2}{2}(g_{t-1} + g_t)^\top H_{t-1}(g_{t-1} + g_t).\end{aligned} \tag{38}$$

The first-order projection can be written as

$$\begin{aligned}&\nabla\mathcal{L}(\theta_{t-2})^\top(g_{t-1} + g_t) \\ &\quad= \|g_{t-1}\|^2 + g_{t-1}^\top g_t - \xi_{t-1}^\top(g_{t-1} + g_t).\end{aligned} \tag{39}$$

Finally, $g_{t-1}^\top g_t = \|g_{t-1}\|\|g_t\|\rho_t$ yields a form of Eq. (38) using consecutive-gradient cosine. The same starting Hessian appears throughout this expansion. For a nonlinear objective, the Taylor remainder is governed by the combined displacement and the variation of the Hessian along it.

Appendix C
Measurement Details and Additional Diagnostics

A. Objectives and Measurement

The tanh MLP has two 200-unit hidden layers and minimizes half-squared error summed over ten output coordinates and averaged over the first 5,000 training images. ResNet-32 minimizes mean cross-entropy on all 50,000 training images. The word-level Transformer has two layers, two attention heads, and embedding/feed-forward dimensions of 200; its objective is token-weighted negative log-likelihood. In the GD comparison, constant learning rates share frozen seed-0 initial tensors within each case. GD uses no momentum, decay, clipping, augmentation, dropout or schedule.

The MLP uses coordinate-wise standardization and ResNet uses channel-wise standardization, both based on the complete CIFAR-10 training split. ResNet's objective averages the same 50 fixed groups of 1,000 images, with batch normalization performed separately within each group. Accumulating all group gradients gives the gradient of this fixed objective; physical batching introduces no random sampling.

For the Transformer, the corpus is arranged into 20 fixed streams and causal chunks of at most 35 tokens. Eight stream-tail tokens and the 20 initial context tokens have no training target; short chunks retain their valid targets, and padding is excluded from the loss. The resulting 59,680 sequences contain 2,088,600 training targets. The 33,278-word vocabulary follows the first-occurrence dictionary over training, validation and test text in the PyTorch word-language-model example, revision `13acec6d7c78` [21]. The same pinned example supplies the Transformer implementation. All reported comparisons concern training-objective dynamics.

Physical accumulation uses 1,000-image chunks for the MLP, the fixed 1,000-image groups for ResNet, and 64-sequence chunks for the Transformer. The complete objective is accumulated before each parameter update. Training parameters and gradients are FP32, TF32 is disabled, and diagnostic reductions are FP64. The nine conditions give paired learning-rate comparisons within three model cases. Architectures, objectives and data coverage differ between cases, and each case has one initialization. The computational cost stated in Section III-C is algorithmic; measured runtime overhead depends on implementation and is left to a separate benchmark.

Cosines with zero norms and the first unpaired gradient are undefined. Complete 50-observation trailing means summarize the GD diagnostic curves; pointwise quantities are computed before averaging. Endpoint losses and R_{3000} are unsmoothed.

B. Unsmoothed Alignment and Temporal Accounting

Figure 9 supplies the unsmoothed consecutive cosines underlying Fig. 2(a)–(c). Complete trailing means suppress short transients and shift the displayed transition relative to the first negative pair. In particular, the MLP at $\eta = 0.04$ reverses near initialization and then returns to positive alignment before its later negative phase. These raw traces distinguish transient regime changes from sustained destructive phases.

For the complete prefix of T gradients, define the traveled distance and the temporal contributions as

$$\begin{aligned} P_T &:= \eta \sum_{t=1}^{T} \|g_t\|, \\ I_T^{(1)} &:= 2 \sum_{t=2}^{T} g_{t-1}^\top g_t, \\ I_T^{(>1)} &:= 2 \sum_{\substack{1 \le u < t \le T \\ t-u > 1}} g_u^\top g_t, \\ I_T &:= I_T^{(1)} + I_T^{(>1)} = \Big\| \sum_{t=1}^{T} g_t \Big\|^2 - \sum_{t=1}^{T} \|g_t\|^2. \end{aligned} \tag{40}$$

Table III reports these quantities at $T = 3000$. The interaction columns use raw gradients and include the factor of two; they are not multiplied by η^2. The total is positive in all nine conditions, whereas the adjacent contribution is negative in seven. The positive total demonstrates longer-range reinforcement alongside adjacent destruction. Magnitudes are compared within each model and its fixed parameterization.

C. Retrospective Loss Approximation

For each gradient pair indexed by $t = 2, \dots, 3000$, let $\delta\mathcal{L}_{t-1} := \mathcal{L}(\theta_{t-1}) - \mathcal{L}(\theta_{t-2})$. We compare this observed change with the gradient-based approximation

$$\widehat{\delta\mathcal{L}}_{t-1} := -\frac{\eta}{2} \|g_{t-1}\|^2 \left(2 - \eta \widehat{\kappa}_t\right). \tag{41}$$

The normalized absolute discrepancy is

$$e_t := \frac{|\delta\mathcal{L}_{t-1} - \widehat{\delta\mathcal{L}}_{t-1}|}{\eta \|g_{t-1}\|^2}. \tag{42}$$

The denominator is the magnitude of the first-order full-GD descent term. It is nonzero for every measured pair. Table IV reports the median over all 2,999 pairs and the count of opposite signs. The final update is excluded because its subsequent gradient was not recorded. All inputs are unsmoothed; the table uses no seed aggregation.

For the threshold sensitivity columns, define

$$h_t(\tau) := \tau \max\{|\mathcal{L}(\theta_{t-2})|, |\mathcal{L}(\theta_{t-1})|\}. \tag{43}$$

A transition is eligible only when both $|\delta\mathcal{L}_{t-1}| > h_t(\tau)$ and $|\widehat{\delta\mathcal{L}}_{t-1}| > h_t(\tau)$. Each cell prints opposite-sign transitions divided by its eligible count; the median discrepancy uses all pairs. The thresholds provide a descriptive sensitivity analysis. The comparison uses the subsequent gradient and repeated transitions within one trajectory; finite-step, nonsmooth and numerical effects all contribute to the discrepancy. For the highest-rate Transformer, the unthresholded signs disagree in 44.2% of transitions; excluding changes below the 10^{-5} relative threshold leaves 43.7% disagreement. Thus a projected change near two identifies the oscillatory regime at the scale shown in the main figures, while stepwise loss signs remain sensitive to nonlinear effects.

D. Backward-Lag Measurements

The backward profiles use the largest GD rate in each model, with exact full-objective gradients from separately recomputed 256-update continuations. Retained states 500, 2000 and 3000 supply common starting parameters; their endpoints are states 756, 2256 and 3256 (gradient indices 757, 2257 and 3257). Figure 3 uses the middle endpoint in every model, providing a common trained-state comparison. Each curve holds that endpoint fixed, includes every lag through 256, and uses no temporal averaging or compression. FP64 products of FP32 vectors were checked independently on CPU and GPU; selected retained vectors supply additional local checks. Later floating-point trajectories can depart from earlier replays despite identical starting tensors, so these profiles are separate continuations, not reconstructed pairs from Figs. 1 and 2. At all nine measured endpoints, both positive and negative similarities occur within lags 193–256; their amplitudes vary substantially across endpoints.

E. Mini-Batch Comparison

The mini-batch comparison uses three initializations for each objective. The MLP reuses the completed 5,000-image experiment at $\eta = 0.02$, with batches $32, 128, 512, 2048, 5000$. Fresh 2,000-update GD references at $\eta = 0.1$ and 0.2 supply ResNet and Transformer starting states. ResNet samples $1, 5, 20$ or all 50 fixed 1,000-image normalization groups, preserving each image's normalization context; its figure axis reports image counts. Transformer batches contain $8, 32, 128, 512$ or all $59{,}680$ prepared sequences, with $M = 2{,}088{,}600$ supervised tokens in the complete training objective. If ℓ_i is sequence i's summed token loss, the sampled estimator is $\frac{N}{bM} \sum_{i \in B} \ell_i$, with $N = 59{,}680$; a random sampled-token denominator would change this estimator. Every stochastic batch is a uniform subset without replacement, independently

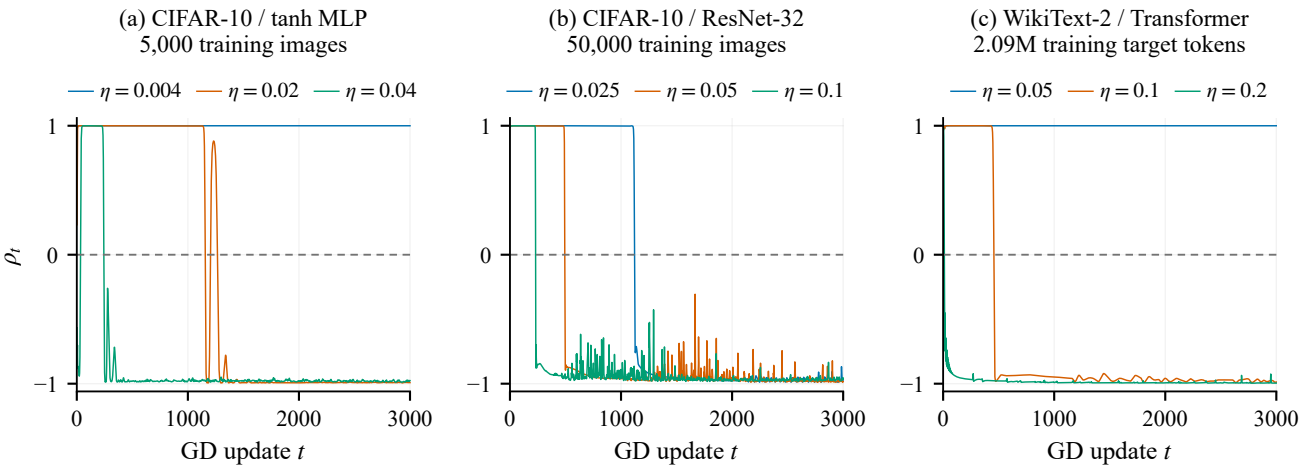


Fig. 9: Unsmoothed consecutive-gradient cosine for the nine full-GD conditions in Section IV, with 2,999 defined pairs per condition. Columns show (a) MLP, (b) ResNet-32 and (c) Transformer. The dashed horizontal reference is zero. Blue, orange and green identify increasing learning rates within each case, as given in the legends. Every curve uses seed 0 and covers the same 3,000-update horizon; the first undefined pair is omitted and there is no temporal or seed averaging. Coincident positive curves can overlap exactly.

redrawn at each update. Batch sizes share prefixes of a common permutation within a repetition; permutations are independent across updates and repetitions. Parameters and gradients use FP32, reductions use FP64, and TF32 is disabled. Full-loss evaluations cover every training image or supervised token. Cosines average all 199 defined consecutive pairs; retention uses all 200 applied gradients. Eight repetitions are averaged within initialization before the three initializations receive equal weight. The plotted range represents variation among initialization-block means. Cases differ in objective, architecture, data size and selected learning rate; the controlled contrasts are within each case.

Cloud runs use PyTorch 2.8/CUDA 12.8. The Transformer uses RTX5090 and 512-sequence physical accumulation, without changing effective batches. ResNet uses L40S/RTX4090; seed 2's smallest two batches use GB10/PyTorch 2.11/CUDA 13 from hash-identical cloud checkpoints. Complete-gradient and short replay checks qualify this runtime; a 200-update initial full-GD bridge has endpoint parameter relative error 2.36×10^{-7} versus the cloud. Batch and runtime effects are confounded in this block; shading includes initialization and runtime variation, and long trajectories can drift beyond the short replay agreement. Initial full controls for ResNet seed 2 and Transformer seed 0 reuse the exact first 200 reference updates, with separate endpoint losses; other controls execute independently. Audits of actual increments and endpoint displacement quantify FP32 update roundoff.

F. Spectral Context

We measure the largest algebraic Hessian eigenvalue at states $0, 100, \ldots, 3000$ using matrix-free Hessian–vector products and a checked relative eigenpair residual of at most 0.005. The Hessian objective covers all 5,000 MLP training examples, but only the first five fixed ResNet groups and the first 2,500 Transformer sequences. Subset measurements follow the computational motivation of [1] and describe sharpness on those fixed subsets. Reused MLP trajectories requested two largest eigenvalues and the other runs requested one; we compare only the largest value. Table V retains all nine final measurements.

Maximum pointwise eigenvalue concerns the most curved local direction, whereas $\widehat{\kappa}_t$ concerns realized gradient change across a step. Objective coverage and a finite-step Taylor remainder also affect their comparison. Final spectral values and the late means in Table I use different temporal reductions. Both coverage and timing must therefore be matched for an estimator-accuracy comparison. A state-matched local comparison would pair the Hessian at state s with the expansion for gradient index $t = s + 2$; the spectrum at state 3000 has no matching recorded gradient pair.

G. Controlled Adam Measurements and Spike Analysis

Adam uses the complete objectives and frozen initial tensors specified above. All trainable parameters enter the diagnostics; parameters, gradients and moments are FP32, reductions are FP64, and TF32 is disabled. All MLP rates use one RTX 5070 Ti. In increasing-rate order, ResNet uses PRO 4000, a PRO 6000 partition and RTX 4090; Transformer uses RTX 4090, PRO 4500 SE and RTX 4090. Initial-gradient agreement is verified across workers. Hardware variation and later finite-precision drift accompany the rate comparison. Held-out splits are evaluated every 128 updates; rate selection uses training calibration.

At state θ_s, the raw gradient is g_{s+1} and the first moment m_{s+1} includes that gradient before update $s + 1$. This is the clock used in the Adam figures. Momentum cosines replace g by m in Eq. (8). Relative norms use the first displayed state, as specified in Fig. 6; event statistics use the fixed pre-rise baseline. All losses and micro-plots are unsmoothed. Initial adjacent diagnostics and zero-denominator ratios are

TABLE III: Complete-prefix temporal accounting at $T = 3000$, with the quantities defined in Eq. (40). P_{3000} is path length; the last three columns are raw-gradient interactions. All nine conditions use one shared initialization per case. Values are rounded only for display, so displayed components may not sum exactly to the displayed total.

Model	η	P_{3000}	$I^{(1)}_{3000}$	$I^{(>1)}_{3000}$	I_{3000}
MLP	0.004	1.1976	1.270×10^{2}	5.634×10^{4}	5.647×10^{4}
MLP	0.02	64.3608	-1.217×10^{4}	3.063×10^{4}	1.846×10^{4}
MLP	0.04	185.5420	-1.662×10^{4}	2.225×10^{4}	5.630×10^{3}
ResNet-32	0.025	41.1785	-2.528×10^{3}	5.325×10^{4}	5.072×10^{4}
ResNet-32	0.05	146.3482	-7.380×10^{3}	3.603×10^{4}	2.865×10^{4}
ResNet-32	0.1	372.7896	-1.076×10^{4}	2.451×10^{4}	1.375×10^{4}
Transformer	0.05	15.6687	1.629×10^{2}	5.100×10^{4}	5.116×10^{4}
Transformer	0.1	111.8465	-7.767×10^{2}	2.464×10^{4}	2.387×10^{4}
Transformer	0.2	405.8404	-2.682×10^{3}	1.336×10^{4}	1.067×10^{4}

TABLE IV: Retrospective check of the local loss approximation for all nine conditions. The unthresholded count and median e_t use 2,999 transitions per condition. Each threshold cell gives opposite signs / eligible transitions; eligibility requires both change magnitudes to exceed $h_t(\tau)$. These descriptive temporal counts are not independent-sample predictive accuracies.

				Opposite signs / eligible transitions at threshold τ		
Model	η	No threshold	Median e_t	10^{-7}	10^{-6}	10^{-5}
MLP	0.004	0/2999	2.905×10^{-4}	0/2999	0/2999	0/2999
MLP	0.02	145/2999	6.146×10^{-4}	145/2998	145/2998	141/2989
MLP	0.04	281/2999	7.343×10^{-3}	281/2999	281/2998	280/2994
ResNet-32	0.025	595/2999	1.030×10^{-2}	595/2999	592/2991	570/2934
ResNet-32	0.05	694/2999	2.347×10^{-2}	694/2999	692/2997	682/2978
ResNet-32	0.1	814/2999	2.324×10^{-2}	813/2997	813/2997	809/2989
Transformer	0.05	0/2999	2.381×10^{-5}	0/2999	0/2999	0/2999
Transformer	0.1	542/2999	1.398×10^{-2}	541/2997	534/2977	436/2703
Transformer	0.2	1325/2999	6.323×10^{-2}	1324/2996	1319/2989	1239/2837

TABLE V: Final spectral context at parameter state 3000. Only the MLP measurement covers its complete training objective. ResNet and Transformer use fixed subsets of the stated size; residual-checked eigenpairs do not establish subset representativeness. All conditions use seed 0.

Model	η	Hessian objective coverage	$\eta\lambda_{\max,3000}$
MLP	0.004	Full 5,000 images	0.3168
MLP	0.02	Full 5,000 images	2.2157
MLP	0.04	Full 5,000 images	2.3259
ResNet-32	0.025	Fixed 5,000-image subset	2.8176
ResNet-32	0.05	Fixed 5,000-image subset	2.6599
ResNet-32	0.1	Fixed 5,000-image subset	3.2488
Transformer	0.05	Fixed 2,500-sequence subset	1.7229
Transformer	0.1	Fixed 2,500-sequence subset	2.2752
Transformer	0.2	Fixed 2,500-sequence subset	3.1202

undefined. The terminal raw-gradient diagnostic is recorded after update 1,024 without applying another update.

Event extraction searches every loss state from initialization, using strict increases and the first return in Eq. (27). The density includes all 806 recovered excursions. The anatomy sample keeps 129 with increase $> 10\%$ and observed recovery; two above-threshold events without recovery by update 1,024 are excluded. Two initialization events lack pre-rise cosine and projected change, leaving 127 scatter points. Each event receives equal weight; recovery summaries first average within the event. Standard deviations use denominator $n - 1$.

The seven scatter fits use ordinary least squares with an intercept, separate coefficients per model/rate, and $\log_{10}(\mathcal{L}_{p_i}/\mathcal{L}_{a_i})$ as response. Reported R^2 uses that response in the fitting sample; dashed extensions are extrapolations. Densities use Gaussian kernels on log percentage increase. The common bandwidth within a model is the pooled sample standard deviation times pooled $n^{-1/5}$, and each rate's density has unit mass in those coordinates. These exploratory event summaries describe repeated observations within one initialization per model. Analysis036 retains source hashes, segmentation checks and scalar histories.

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